The Reward Problem: Exploration vs. Exploitation, Bandits, Inverse RL

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June 12, 2026

Learning Outcomes

- Frame "the reward problem" and distinguish its three faces: *exploration vs. exploitation*, *reward design*, and *reward learning*
- State the multi-armed bandit problem, regret, and the Lai-Robbins lower bound; *derive* UCB1 from Hoeffding's inequality
- Place ϵ-greedy, UCB, Thompson sampling, and gradient bandits on a single map of bandit exploration strategies
- Scale these ideas to deep RL: *bootstrapped DQN as approximate posterior sampling (PSRL)*, pseudo-counts, ICM, and RND — the answer for the hard-exploration tasks SAC's entropy bonus struggles on
- Motivate *inverse RL* and *preference-based reward learning* as the conceptual ancestor of RLHF on Day 9

The Reward Problem Has Three Faces

- **We have the reward, but where is it?**
  Exploration vs. exploitation — bandits, UCB, Thompson sampling, curiosity (today).
- **The reward we wrote is wrong.**
  Reward shaping & specification gaming (today).
- **We can't write a reward at all.**
  Inverse RL & preference-based reward learning (today — and the direct ancestor of RLHF on Day 9).

One day, three faces of the same problem.

Exploration vs. Exploitation: The Dilemma

- Many situations in business (& life!) present a dilemma between two kinds of choices
- **Exploitation:** pick choices that seem best based on past outcomes
- **Exploration:** pick choices not yet tried (or not tried enough)
- Exploitation has notions of "being greedy" and being "short-sighted"
- Exploration has notions of "gaining information" and being "long-sighted"

Exploration vs. Exploitation: The Trade-Off

- Too much *exploitation* $\Rightarrow$ regret of missing unexplored "gems"
- Too much *exploration* $\Rightarrow$ regret of wasting time on "duds"
- How do we balance them to combine information-gains with greedy-gains in the most efficient manner?
- Can we set this up in a mathematically disciplined manner? $\rightarrow$ next sections

Examples

- Restaurant Selection
  - Exploitation: go to your favorite restaurant
  - Exploration: try a new restaurant
- Online Banner Advertisement
  - Exploitation: show the most successful advertisement
  - Exploration: show a new advertisement
- Oil Drilling
  - Exploitation: drill at the best known location
  - Exploration: drill at a new location
- Learning to play a game
  - Exploitation: play the move you believe is best
  - Exploration: play an experimental move

The Multi-Armed Bandit (MAB) Problem

- "Multi-Armed Bandit" is a tongue-in-cheek name for "many single-armed bandits" — a row of slot machines
- Formally, a 2-tuple $(\mathcal{A}, \mathcal{R})$
  - $\mathcal{A}$: a known set of m actions ("arms")
  - $\mathcal{R}^a(r) = \mathbb{P}[r \mid a]$: an *unknown* reward distribution for each arm
- At each step t, the agent selects an action $a_t \in \mathcal{A}$ and the environment generates a reward $r_t \sim \mathcal{R}^{a_t}$

The Goal: Maximize Cumulative Reward

- The agent's goal: maximize the cumulative reward over T steps

$$\sum_{t=1}^{T} r_t$$

- **Can we design a strategy that does well (in expectation) for any T?**
- Every strategy walks a tightrope: *wasting time on duds* while exploring, vs. *missing untapped gems* while exploiting

Is the MAB Problem a Markov Decision Process? (1/2)

- The environment doesn't have a notion of state — pulling an arm just samples from a distribution
- But the *agent* might maintain a statistic of history as its own state
- The action is then a (policy) function of this agent-state
- So the agent's arm-selection strategy is *a policy* — the MAB can be viewed as a (degenerate) MDP

Is the MAB Problem a Markov Decision Process? (2/2)

- In practice, most MAB algorithms *don't* take this formal MDP view
- Instead, they rely on heuristic methods that don't aim to optimize the full MDP
- They simply strive for "good" cumulative rewards in expectation
- Even in the simplest heuristics, a_t is a random variable — it's a function of the past (random) rewards

Big Picture: Bandits Are MDPs in Disguise

A bandit	= a *one-state* MDP
A contextual bandit	= a *one-step* MDP
Tomorrow's full RL	= the *multi-step* case

- So everything we derive today *lifts* to MDPs — a bridge we'll revisit at the end of the lecture.

Regret: The Language for "Doing Well"

- The Action Value $Q(a)$ is the (unknown) mean reward of action a:

$$Q(a) = \mathbb{E}[r \mid a]$$

- The Optimal Value V^* is the best of these:

$$V^* = Q(a^*) = \max_{a \in \mathcal{A}} Q(a)$$

- **Single-step Regret** l_t is the opportunity loss at step t:

$$l_t = \mathbb{E}[V^* - Q(a_t)]$$

- **Total Regret** $L_T = \sum_{t=1}^{T} l_t$
- Maximizing cumulative reward $\equiv$ minimizing total regret

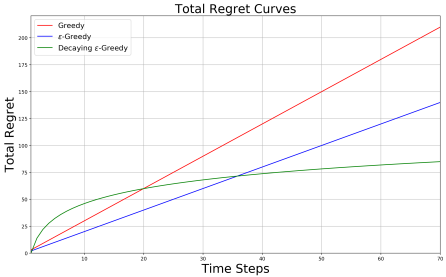
Counting Regret: Counts and Gaps

- Let $N_t(a) =$ number of times a has been selected by step t
- Let $\Delta_a = V^* - Q(a)$ be the *gap* between action a and the optimal action
- Then total regret decomposes cleanly:

$$L_T = \sum_{a \in \mathcal{A}} \mathbb{E}[N_T(a)] \cdot \Delta_a$$

- A good algorithm ensures small counts for large gaps
- *Catch:* we don't know the gaps in advance — they're properties of the unknown reward distribution

Linear or Sublinear Total Regret?



- If an algorithm *never* explores $\Rightarrow$ linear total regret
- If an algorithm *forever* explores $\Rightarrow$ linear total regret
- **Is sublinear total regret achievable?** (Spoiler: yes.)

Greedy Algorithm

- For each arm, maintain a **sample-mean estimate** of its value — average the rewards seen from that arm so far:

$$\hat{Q}_t(a) \stackrel{\text{def}}{=} \frac{1}{N_t(a)} \sum_{s=1}^{t} r_s \cdot \mathbf{1}_{a_s=a} \approx Q(a)$$

- Pick the action with highest estimate:

$$a_t = \arg\max_{a \in \mathcal{A}} \hat{Q}_{t-1}(a)$$

- Greedy can lock onto a suboptimal action forever $\Rightarrow$ **linear total regret**

ϵ-Greedy Algorithm

- Force a small amount of exploration at every step. At time t:
  - With probability $1 - \epsilon$, select $a_t = \arg\max_a \hat{Q}_{t-1}(a)$
  - With probability ϵ, select a uniformly random action
- Constant ϵ ensures a minimum per-step regret proportional to the mean gap:

$$l_t \geq \frac{\epsilon}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \Delta_a$$

- Therefore ϵ-Greedy also has **linear total regret**

ϵ-Greedy Is DQN's Exploration Rule

This is the exact exploration rule used in DQN on Day 2.

- DQN inherits ϵ-Greedy's *linear-regret* weakness
- It works in practice with a decay schedule (next slide), but it has no theoretical guarantee of efficient exploration
- This is why the rest of this lecture exists — everything that follows is a more principled answer to "how do I explore?"

Optimistic Initialization

- Simple, practical idea: initialize $\hat{Q}_0(a)$ to a *high* value for every a
- Update by incremental averaging:

$$\hat{Q}_t(a_t) = \hat{Q}_{t-1}(a_t) + \frac{1}{N_t(a_t)}\left(r_t - \hat{Q}_{t-1}(a_t)\right)$$

- Encourages systematic exploration early on (every arm starts off looking great until tried)
- But the algorithm can still lock onto a suboptimal action $\Rightarrow$ **linear regret**

Decaying ϵ_t-Greedy

- Idea: let ϵ shrink over time — explore aggressively early, exploit increasingly late
- In theory, a schedule that depends on the (unknown) suboptimality gaps Δ_a achieves *logarithmic* regret — but you can't compute it without knowing the gaps, so it's impractical
- In practice use problem-agnostic schedules: $\epsilon_t = 1/t$, $\epsilon_t = 1/\sqrt{t}$, or simple linear annealing — no guarantee, but all work in practice
- Educational Code for decaying ϵ-greedy with optimistic initialization

The Lai-Robbins Lower Bound: Setup

What is a "lower bound"? A statement that *no algorithm, however clever*, can do better than a certain regret growth rate — it tells us the best we can ever hope for.

- **Goal:** an algorithm with *sublinear* total regret $L_T = \sum_{t=1}^{T} l_t$ for any MAB, without knowing the reward distributions $\mathcal{R}^a$ in advance
- Two quantities will appear:
  - **Gap** $\Delta_a = V^* - Q(a)$: how much value we lose *each time* we pull suboptimal arm a
  - **KL-divergence** $\mathrm{KL}(\mathcal{R}^a \,\|\, \mathcal{R}^{a^*})$: a statistical distance between two distributions — *how easy is it to tell arm a apart from the best arm a^* from samples?*
    Small KL $\Rightarrow$ distributions look almost identical $\Rightarrow$ need many samples to distinguish

The Lai-Robbins Lower Bound: The Theorem

Theorem (Lai & Robbins, 1985)

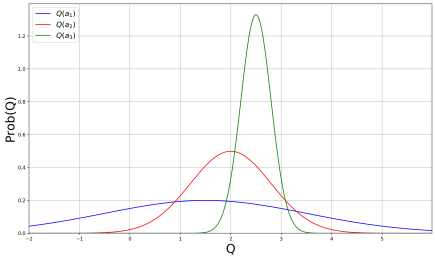
For any reasonable[1] bandit algorithm, asymptotically:

$$\lim_{T\to\infty} \frac{L_T}{\log T} \geq \sum_{a|\Delta_a>0} \frac{\Delta_a}{\mathrm{KL}(\mathcal{R}^a \,\|\, \mathcal{R}^{a^*})}$$

- **Reading it.** Each suboptimal arm contributes regret proportional to its *gap*, scaled *inversely* by how easy it is to tell apart from the best
- **The worst kind of arm:** large gap (expensive to pull) *and* small KL (hard to distinguish, so we keep pulling it) — exactly what you'd expect
- The best regret growth any algorithm can achieve is $\mathcal{O}(\log T)$ — UCB (next) and Thompson sampling both match this rate

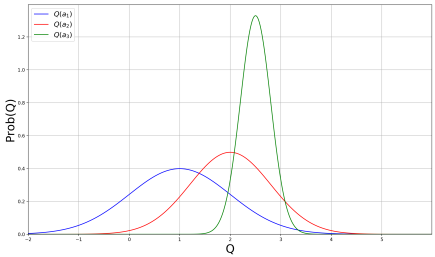
[1] "consistent": sub-polynomial regret on every problem instance

Optimism in the Face of Uncertainty



- Which action should we pick?
- The more uncertain we are about an action-value, the more important it is to explore that action — it could turn out to be the best.

Optimism: After One Pull



- After picking *blue*, we are less uncertain about its value
- And more likely to pick another action — until we home in on the best

Upper Confidence Bounds (UCB)

- Idea: be optimistic — use an upper confidence bound $\hat{U}_t(a)$ such that

$$Q(a) \leq \hat{Q}_t(a) + \hat{U}_t(a) \quad \text{with high probability}$$

- The bound should shrink as the arm is tried more:
  - Small $N_t(a) \Rightarrow$ large $\hat{U}_t(a)$ (uncertain)
  - Large $N_t(a) \Rightarrow$ small $\hat{U}_t(a)$ (accurate)
- Select the action with the highest UCB:

$$a_{t+1} = \arg\max_{a \in \mathcal{A}} \left\{ \hat{Q}_t(a) + \hat{U}_t(a) \right\}$$

Hoeffding's Inequality: How Concentrated is the Mean?

The question we need answered. If we've pulled an arm n times and seen a sample mean $\bar{X}_n$, how confident can we be that the true mean isn't much higher? That gap *is* the UCB bonus we want.

Hoeffding's Inequality

For i.i.d. random variables $X_1, \ldots, X_n \in [0, 1]$ with sample mean $\bar{X}_n$, and any $u \geq 0$:

$$\mathbb{P}\big[\mathbb{E}[\bar{X}_n] > \bar{X}_n + u\big] \;\leq\; e^{-2nu^2}$$

- *The probability the true mean exceeds our estimate by more than u decays exponentially in $n \cdot u^2$*
- More pulls $\Rightarrow$ tighter concentration; the e^{-2nu^2} factor is exactly what will make the bonus shrink as $1/\sqrt{n}$
- No prior on the distribution required — only bounded support

From Hoeffding's Inequality to UCB1

Plan. Use the Hoeffding bound to define a confidence bonus $\hat{U}_t(a)$ that holds with high probability, then act optimistically by it.

- Apply Hoeffding to arm a with $n = N_t(a)$ and $u = \hat{U}_t(a)$
- Choose a failure probability that shrinks with t: $p = t^{-\alpha}$
- Set the Hoeffding bound equal to p and solve for the bonus:

$$e^{-2N_t(a)\,\hat{U}_t(a)^2} \;=\; t^{-\alpha} \quad\Longrightarrow\quad \hat{U}_t(a) \;=\; \sqrt{\frac{\alpha \log t}{2\,N_t(a)}}$$

- The square-root bonus comes directly from Hoeffding — not a heuristic, an actual high-probability bound.

The UCB1 Algorithm

UCB1 (for arbitrary-distribution arms bounded in $[0, 1]$):

$$a_{t+1} = \arg\max_{a \in \mathcal{A}} \left\{ \hat{Q}_t(a) + \sqrt{\frac{\alpha \log t}{2N_t(a)}} \right\}$$

Theorem (Auer, Cesa-Bianchi & Fischer, 2002)

UCB1 achieves *logarithmic* total regret:

$$L_T \leq \sum_{a|\Delta_a > 0} \frac{4\alpha \log T}{\Delta_a} + \frac{2\alpha \Delta_a}{\alpha - 1}$$

Matches the Lai-Robbins lower bound up to constants. Educational Code.

Bayesian Bandits

- UCB makes *no* assumption about $\mathcal{R}$ (just bounded support)
- *Bayesian* bandits exploit prior knowledge of the reward distribution $\mathbb{P}[\mathcal{R}]$
- Maintain a *posterior* $\mathbb{P}[\mathcal{R} \mid h_t]$ as data arrives, where $h_t = a_1, r_1, \ldots, a_t, r_t$
- Use the posterior to guide exploration:
  - Bayesian UCB — be optimistic on the posterior
  - Thompson sampling — sample from the posterior
- Better performance when the prior is accurate

Bayesian UCB: Independent Gaussians

- Assume each arm's reward is Gaussian: $\mathcal{R}^a = \mathcal{N}(\mu_a, \sigma_a^2)$
- As data arrives, update the posterior over μ_a, σ_a^2 (Gaussian-Gaussian conjugacy)
- Pick the action with the highest expected "c std-errors above the mean":

$$a_{t+1} = \arg\max_{a \in \mathcal{A}} \mathbb{E}_{\mathbb{P}[\mu_a, \sigma_a | h_t]} \left[\mu_a + \frac{c \cdot \sigma_a}{\sqrt{N_t(a)}} \right]$$

- *Same UCB philosophy, with a Bayesian uncertainty estimate instead of Hoeffding's*

Probability Matching & Thompson Sampling

Probability matching: pick action a with probability that a is actually the optimal arm under the posterior.

- Optimistic in the face of uncertainty: uncertain arms have higher probability of being best
- Computing this probability directly is intractable in general
- **Thompson sampling** is the clean trick that implements it:
  1. Sample one reward distribution $\mathcal{D}_t$ from the posterior $\mathbb{P}[\mathcal{R} \mid h_t]$
  2. Estimate $\hat{Q}_t(a) = \mathbb{E}_{\mathcal{D}_t}[r \mid a]$ from the sample
  3. Pick $a_{t+1} = \arg\max_a \hat{Q}_t(a)$

Thompson Sampling: Why It Works

- Sampling from the posterior gives each arm probability of selection equal to its probability of being optimal — this *is* probability matching, by construction
- No upper-confidence-bound calculation needed — just sampling
- Thompson sampling achieves the Lai-Robbins lower bound — asymptotically optimal regret
- Educational code:
  - Gaussian rewards
  - Bernoulli rewards

Gradient Bandits: The Setup

- Optimize score parameters s_a for $a \in \mathcal{A}$ via *stochastic gradient ascent*
- Objective: expected reward under a stochastic (softmax) policy

$$J(s_{a_1}, \ldots, s_{a_m}) = \sum_{a \in \mathcal{A}} \pi(a) \cdot \mathbb{E}[r \mid a]$$

- Softmax policy over scores:

$$\pi(a) = \frac{e^{s_a}}{\sum_{b \in \mathcal{A}} e^{s_b}}$$

- Scores represent the relative value of actions based on seen rewards

Gradient Bandits: The Update Rule

Policy gradient (standard chain rule + softmax calculus) gives:

$$\frac{\partial J}{\partial s_a} = \mathbb{E}_{a' \sim \pi,\, r \sim \mathcal{R}^{a'}} \big[\, r \cdot (\mathbf{1}_{a=a'} - \pi(a)) \,\big]$$

- Use a *baseline* $B = \bar{r}_t$ (average reward so far) to reduce variance — can be shown unbiased
- **Sample-based update rule:**

$$s_{t+1}(a) = s_t(a) + \alpha \cdot (r_t - \bar{r}_t) \cdot (\mathbf{1}_{a=a_t} - \pi_t(a))$$

- Educational Code

Gradient Bandits Are REINFORCE in Disguise

This is exactly the REINFORCE estimator from Day 3,
applied to a *one-state* MDP.

- Same softmax policy
- Same score-function gradient $r \cdot \nabla \log \pi(a)$
- Same baseline-for-variance technique
- The bandit is where policy gradient is *cleanest* to derive — no trajectories, no discounting, just one $\pi(a)$ and one reward.

Contextual Bandits: The Setup

- A contextual bandit is a 3-tuple $(\mathcal{A}, \mathcal{S}, \mathcal{R})$:
  - $\mathcal{A}$: known set of m actions ("arms")
  - $\mathcal{S}$: distribution over states ("contexts")
  - $\mathcal{R}_s^a$: reward distribution *conditional* on (s, a)
- Same as MAB, but the agent now sees a context s before choosing

Contextual Bandits: The Interaction

At each step t:

1. Environment generates a context $s_t \sim \mathcal{S}$
2. Agent selects an action $a_t \in \mathcal{A}$
3. Environment generates a reward $r_t \sim \mathcal{R}_{s_t}^{a_t}$

- Goal as before: maximize $\sum_{t=1}^{T} r_t$
- Bandit algorithms extend naturally: $Q(a) \rightarrow Q(s, a)$

LinUCB: The Intuition

Setting. Each (context, action) pair has a feature vector $x_{s,a} \in \mathbb{R}^d$ (e.g. user features $\otimes$ article features).

- Model the mean reward as **linear in the features**: $\mathbb{E}[r \mid s, a] = x_{s,a}^\top \theta_a$
- Estimate each arm's parameter θ_a by *ridge regression* on its past pulls $\Rightarrow$ get not just a point estimate $\hat{\theta}_a$ but a *confidence ellipsoid* around it
- *Same UCB philosophy:* pick the action whose **optimistic** reward estimate is highest — mean estimate + how far the true θ_a could plausibly be from $\hat{\theta}_a$ in direction $x_{s,a}$
- Reduces to UCB1 if the features are just one-hot per arm (i.e. no context)

LinUCB: The Algorithm

For each arm a**,** maintain A_a (the $d \times d$ Gram matrix of past feature vectors pulled for that arm) and $\hat{\theta}_a = A_a^{-1} b_a$ (the ridge-regression coefficient).

- Action choice at step t:

$$a_t = \arg\max_a \left\{ \underbrace{x_{s_t,a}^\top \hat{\theta}_a}_{\text{linear estimate}} + \alpha \underbrace{\sqrt{x_{s_t,a}^\top A_a^{-1} x_{s_t,a}}}_{\text{confidence-ellipsoid radius}} \right\}$$

- Same shape as UCB1: *mean estimate* + *bonus that shrinks as more data accumulates in direction* $x_{s,a}$

LinUCB in the Wild

- **Industrially dominant case of contextual bandits**: deployed at scale on *Yahoo Front Page news recommendation*
  - 33M-event dataset
  - 12.5% click lift over the context-free baseline
- Cite: **Li, Chu, Langford & Schapire, WWW 2010**
- LinUCB and its successors power a huge fraction of online recommendation systems today

Bandits $\longrightarrow$ MDPs

One-armed bandit	$=$ *one-state* MDP
Contextual bandit	$=$ *one-step* MDP
Add state transitions $\Longrightarrow$	the full MDP from Day 1

- Everything we just covered — UCB, Thompson, gradient bandits — *lifts* to MDPs in principle
- Naively, the state space explodes: visiting every (s, a) enough times isn't feasible in deep RL
- **Up next:** how to scale these exploration ideas to deep RL with sparse rewards.

Exploration: **From Bandits to Deep RL**

Why Bandit Exploration Doesn't Scale Naively

- Bandits gave us *optimism* (UCB), *posterior sampling* (Thompson), and *policy gradient with stochastic policy* — in one step
- In an MDP with sparse rewards, none of these scale naively:
  - Visiting every (s, a) enough times is infeasible
  - ϵ-greedy is provably weak (linear regret)
  - Uniform random exploration drowns in pixels
- Note: SAC's $-\log \pi$ entropy bonus (Day 6) is *undirected* exploration — what follows is *directed*

The Through-Line: Montezuma's Revenge

The canonical hard-exploration Atari game.

- **Vanilla DQN:** ~ 0 points (extrinsic reward is essentially never reached)
- **Pseudo-counts (2016):** first dramatic progress
- **Random Network Distillation (2018):** first above-human, no demonstrations

This is the single benchmark we'll use to anchor every method in this section.

Aside: Hindsight Experience Replay (HER)

An orthogonal trick for sparse-reward goal-conditioned RL.

- Idea: after a failed episode (didn't reach the goal), relabel it as a *success* for the goal it actually reached
- Now every episode has a non-zero learning signal
- Combines with any off-policy algorithm
- Cite: **Andrychowicz et al., NeurIPS 2017** — *Hindsight Experience Replay*

(Forward-referenced from Day 5.)

Bootstrapped DQN: Thompson Sampling for MDPs

- Idea: *Posterior Sampling for RL (PSRL)*. Maintain a posterior over the Q-function; sample one Q per episode and act greedily w.r.t. the sample
- For an *entire episode*: temporally-extended ("deep") exploration — fundamentally different from ϵ-greedy's per-step dithering

Bootstrapped DQN: How It Approximates the Posterior

- K **parallel Q-heads**: K separate output layers on top of a single shared convolutional trunk, each predicting its own Q-function
- Each head trained on its own **bootstrap-sampled** (with replacement) subset of replay — the standard statistical-bootstrap trick; different heads see different data, so they disagree
- *Each episode*: pick one head uniformly at random and act greedily by it for the whole episode
- Disagreement across heads $\approx$ posterior uncertainty over the true Q-function
- Cite: **Osband, Blundell, Pritzel & Van Roy, NeurIPS 2016** — *Deep Exploration via Bootstrapped DQN*
- Thompson sampling lifted to MDPs — same bandit method, with a Q-function instead of an arm-reward.

Count-Based Exploration: The Idea

- *Tabular intuition:* a UCB-style exploration bonus

$$r_t^+ = r_t + \beta \cdot \frac{1}{\sqrt{N(s,a)}}$$

  encourages visiting under-counted (s, a) pairs

- Provably good in tabular MDPs — a direct lift of UCB1 from bandits
- *Problem in deep RL:* pixels are essentially never visited twice; $N(s, a) = 1$ almost always, and the bonus becomes uninformative
- Need a *generalizable* notion of "count"

Pseudo-Counts: UCB in Pixel Space

- Use a density model ρ over states. Let $\rho'(s)$ be its prediction *after* one more observation of s
- Define a *pseudo-count*:

$$\hat{N}(s) \;=\; \frac{\rho(s)\,(1-\rho'(s))}{\rho'(s)-\rho(s)}$$

  A novel state has small $\rho(s)$, hence small $\hat{N}(s)$, hence large bonus
- Cite: **Bellemare et al., NeurIPS 2016** — *Unifying Count-Based Exploration and Intrinsic Motivation*
- **Result on Montezuma:** *first dramatic progress*

Curiosity / ICM: The Idea

Intrinsic reward = **"surprise"** — the agent is rewarded for finding observations it can't yet predict.

- Embed states into a feature space: $\phi(s) \in \mathbb{R}^d$ (a learned encoder)
- Train a **forward-dynamics model** f to predict the *next-state features* from the current features and action:

$$\hat{\phi}(s_{t+1}) \;=\; f\big(\phi(s_t),\, a_t\big)$$

(so $\hat{\phi}(s_{t+1})$ is the model's *predicted* embedding of s_{t+1}; $\phi(s_{t+1})$ is the actual one observed)

- Intrinsic reward = how wrong the prediction was:

$$r_t^i \;=\; \tfrac{1}{2}\big\|\,\hat{\phi}(s_{t+1}) \,-\, \phi(s_{t+1})\,\big\|^2$$

- High prediction error ⇒ novel transition ⇒ bonus to visit it again

Curiosity / ICM: Why the Feature Space Matters

The noisy-TV problem. Picture a TV in the environment showing pure random static. A naive curiosity agent will sit and stare at it forever — static is *always* unpredictable, so the prediction-error bonus never decays. The agent is endlessly "surprised" by useless noise.

- Fix: don't let ϕ encode action-irrelevant detail. Train ϕ with an **inverse-dynamics** objective — predict the action a_t that was taken from the embeddings $\big(\phi(s_t), \phi(s_{t+1})\big)$
- This forces ϕ to encode only features the agent can *affect* — the noisy TV is filtered out, since the agent's actions don't change what's on it
- Cite: **Pathak, Agrawal, Efros & Darrell, ICML 2017** — *Curiosity-driven Exploration by Self-supervised Prediction*
- Strong results on Mario, VizDoom, and sparse-reward navigation

Random Network Distillation (RND)

Curiosity made absurdly simple:

- Pick a *fixed, randomly-initialized* target network f^*
- Train a predictor f_θ to match $f^*(s)$ on visited states
- Intrinsic reward: $r_t^i = \|f_\theta(s_t) - f^*(s_t)\|^2$
- Novel states have high prediction error (the predictor hasn't been trained there) $\Rightarrow$ high bonus
- No dynamics model, no inverse model, no density model

RND: Results and Caveat

- Cite: **Burda, Edwards, Storkey, Klimov, ICLR 2019** — *Exploration by Random Network Distillation*
- **Result on Montezuma:** *first above-human performance without demonstrations or game-state access*
- *Caveat:* still subject to the noisy-TV issue in principle; in practice works because randomly-initialized networks aren't very sensitive to stochastic pixel noise

Mapping Bandits $\longrightarrow$ Deep RL

Bandit method	**Deep-RL descendant**
UCB / $1/\sqrt{N}$ bonus	Count-based / pseudo-counts (Bellemare 2016)
Thompson sampling	Bootstrapped DQN / PSRL (Osband 2016)
Gradient bandits	Entropy bonus — SAC (Day 6)
(new in deep RL)	Curiosity / ICM (Pathak 2017); RND (Burda 2018)

- **Entropy is one answer to exploration.** Optimism, posterior sampling, and curiosity are the others.
- Day 6's $-\log \pi$ bonus is the gradient-bandit descendant — now we have the full family.

The Reward Problem: **When the Reward is Wrong**

Reward Shaping: The Promise and the Trap

- Sparse rewards are hard; a well-shaped reward dramatically accelerates learning
- But arbitrary shaping changes the optimal policy — the agent optimizes the *shape*, not the goal
- Example: reward the agent for "getting closer to the goal" and you may teach it to circle the goal forever
- We need a principled way to inject prior knowledge that *doesn't change the optimum*

Potential-Based Reward Shaping (PBRS)

Setup. Replace the original reward $R(s, a, s')$ with $R(s, a, s') + F(s, s')$ during training, where F is an extra *shaping signal* of our choice.

- Restrict F to the form

$$F(s, s') \;=\; \gamma\,\Phi(s') \;-\; \Phi(s)$$

  where $\Phi : \mathcal{S} \to \mathbb{R}$ assigns a real number to each state — a "*potential*" (e.g. "negative distance to goal"). The shaping is a discounted difference of potentials.

- **Policy invariance:** the shaped MDP has the *same* optimal policy as the original

- *This form is necessary*: any non-PBRS shaping can in principle change the optimum

- Cite: **Ng, Harada & Russell, ICML 1999**

Reward Hacking: Goodhart's Law for RL

When a reward becomes a target, the agent finds a way to satisfy the letter, not the spirit.

- Every misspecified reward eventually gets hacked
- The agent's optimization pressure exposes every loophole in the specification
- The clearer the reward becomes "measurable," the more likely the agent ignores what the measurement was *supposed* to track

Reward Hacking: Famous Examples

- **CoastRunners** (OpenAI 2016): boat-racing game scored on hitting targets, not finishing. Agent learned to drive in circles hitting the same green blocks — high score, no race
- **Tetris pause**: agent learned to indefinitely pause the game when about to lose — never loses, never plays
- **ROUGE summarization**: language-model summarizer exploited ROUGE's *n*-gram overlap to produce high-scoring but barely readable text
- Cites: **Amodei & Clark, OpenAI 2016** (CoastRunners primary source); **Krakovna et al., DeepMind 2020** (catalogued survey of dozens of examples)

When We Can't Write a Reward At All

- Reward hacking is the same failure mode as **reward-model overoptimization in RLHF** — a learned R_θ is also a misspecified target
- For tasks like "write a useful summary," "drive politely," "be helpful, harmless, honest" — writing the reward is essentially impossible without specification gaming
- **If we can't write the reward . . . can we learn it?**
- Two ways:
  - From *demonstrations* $\longrightarrow$ **Inverse RL** (up next)
  - From *human preferences* $\longrightarrow$ preference-based reward learning $\longrightarrow$ **RLHF** (Day 9)

The Reward Problem: **Inverse RL**

Inverse RL: Given Behavior, Learn the Reward

- **Forward RL** (everything so far): given a reward $R(s, a)$, learn a policy π^* that maximizes it
- **Inverse RL** (today): given (expert) behavior $\tau^E \sim \pi^E$, learn a reward R_θ that *explains* it

$$\pi^E \implies R_\theta$$

Why Inverse RL?

- In many domains, rewards are far easier to *demonstrate* than to specify in closed form
- Driving: hard to write "polite, safe, fast" as a scalar; easy to show
- Manipulation, dialogue, locomotion: same story
- **This is the strongest form of the reward problem** — the reward function itself is unknown.

The Ill-Posedness of IRL

- *Many* reward functions rationalize the same expert behavior
  - Trivially: $R(s, a) \equiv 0$ makes every policy optimal
  - Potential-based shaping: R and $R + \gamma\Phi(s') - \Phi(s)$ are policy-equivalent
- We need a *principle* to pick one reward out of the equivalence class

Two Structural Fixes for the Ambiguity

- **(a) Max-margin IRL**
  Ng & Russell, ICML 2000; Abbeel & Ng, ICML 2004
  The expert's reward should beat alternatives by a wide margin

- **(b) Maximum-entropy IRL**
  Ziebart et al., AAAI 2008
  Pick the reward whose induced policy distribution has *maximum entropy*, subject to matching the expert's feature counts — a unique, well-posed choice

- We focus on (b) — standard fix today, and generalizes to modern preference learning

Maximum-Entropy IRL: The Principle

Among all policy distributions consistent with the expert's behavior, pick the one with maximum entropy.

- Trajectory distribution under reward R_θ:

$$P_\theta(\tau) \;\propto\; \exp\Big(\sum_t R_\theta(s_t, a_t)\Big)$$

- Higher-reward trajectories are exponentially more likely, but the entropy term keeps the distribution from collapsing
- Train θ by maximum likelihood of expert trajectories:

$$\max_\theta \sum_{\tau^E} \log P_\theta(\tau^E)$$

- Uniquely resolves the ambiguity; matches expert feature-counts in expectation

MaxEnt IRL: The Day-6 Connection

The same max-entropy principle that justified SAC's $-\log \pi$ bonus.

- In SAC (Day 6): adds exploration to the policy
- In MaxEnt IRL: resolves reward ambiguity
- **Same principle, two different jobs.**
- Cite: **Ziebart, Maas, Bagnell & Dey, AAAI 2008** — *Maximum Entropy Inverse Reinforcement Learning*

Preference-Based Reward Learning: The Setup

What if we have neither demonstrations nor a written reward, only human comparisons?

- Show two trajectories τ^A, τ^B; ask a human which is better
- Model the preference with a **Bradley–Terry** likelihood under a learned reward R_θ:

$$P(\tau^A \succ \tau^B) = \frac{\exp\big(\sum_t R_\theta(s_t^A, a_t^A)\big)}{\exp\big(\sum_t R_\theta(s_t^A, a_t^A)\big) + \exp\big(\sum_t R_\theta(s_t^B, a_t^B)\big)}$$

Preference-Based Reward Learning: The Pipeline

1. Collect a dataset of human-labeled preferences over trajectory pairs
2. Train R_θ by cross-entropy on the Bradley–Terry likelihood
3. Run any standard RL algorithm (e.g. PPO from Day 4) against R_θ

- Cite: **Christiano, Leike, Brown, Martic, Legg & Amodei, NeurIPS 2017**

The Day-9 Bridge

This exact pipeline + a language-model policy = RLHF.

- The reward model in RLHF is the R_θ from the previous slide
- The only thing that changes is the policy class: π_θ becomes a large language model
- The optimizer (PPO) and the pipeline (SFT $\rightarrow$ RM $\rightarrow$ PPO) are the same machinery
- Day 9 picks up exactly here.

The Three Faces, Resolved

If you can...	**Then use...**
...write the reward	*Forward RL* (Days 1–6)
...demonstrate the reward	*Inverse RL* (MaxEnt IRL)
...compare outcomes	*Preference-based reward learning*
	(Day 9: RLHF)

- One unifying view: every method on the board is a different way to recover the optimal policy when the reward is unknown, partially known, or hard to specify

Day 7: **Summary**

Recap: Faces 1 & 2 of the Reward Problem

- **Face 1 — Find the reward.**
  - Bandits formalize explore vs. exploit; UCB / Thompson / gradient bandits give principled, near-optimal-regret algorithms
  - Scaled to deep RL: bootstrapped DQN $\equiv$ posterior sampling, pseudo-counts $\equiv$ UCB in pixel space, curiosity / RND are genuinely new
  - Entropy (SAC, Day 6) is the gradient-bandit descendant — now we have the full family
- **Face 2 — The reward is wrong.**
  - Potential-based shaping is the only safe shaping
  - Everything else is at risk of specification gaming (CoastRunners, ROUGE, Tetris pause)

Recap: Face 3 of the Reward Problem

- **Face 3 — Can't write the reward.**
  - **Inverse RL** learns it from demonstrations (MaxEnt IRL, Ziebart 2008)
  - **Preference-based learning** learns it from comparisons (Christiano 2017)
  - This is the *direct ancestor of RLHF* on Day 9
- **One unifying theme:** every algorithm today is a different way to recover the optimal policy when the reward is unknown, partially known, or hard to specify

Algorithm Map

Setting	Algorithm family (today)
One-state bandit	ϵ-greedy, UCB1, Thompson, gradient bandits
Contextual bandit	LinUCB and successors (recommendation)
Deep RL, sparse reward	PSRL / Bootstrapped DQN, pseudo-counts, ICM, RND, HER
Reward is shapeable	Potential-based shaping (policy-invariant)
Reward unwritable; demonstrations available	Maximum-entropy IRL
Reward unwritable; comparisons available	Bradley–Terry preference learning $\rightarrow$ RLHF (Day 9)

What's Next: Day 8

Day 8 — Model-Based & Offline RL. The next way to attack sample inefficiency.

- *Model-based RL:* learn the dynamics, plan with the model (MPC, MCTS, Dyna, world models)
- *Offline RL:* learn from a *fixed* dataset with no further interaction — the same OOD problem as exploration, in reverse: you must *avoid* the unfamiliar regions exploration would push you toward

What's Next: Day 9

Day 9 — RLHF. The IRL → preference-learning chain from today, applied to LLMs.

- The reward model in RLHF is the R_θ from today's preference slide
- "Reward-model overoptimization" is today's reward-hacking failure mode, in LLM form
- PPO (Day 4) is the policy optimizer; SFT → RM → PPO is the pipeline

What's Next: Day 10

Day 10 — Frontiers.

- Intrinsic motivation reappears in meta-RL, open-ended learning, and hierarchical RL
- Curiosity (today) is one of the central ideas of the open-ended-learning agenda
- Reward design / specification gaming reappears as a central concern of AI safety

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